

§ Lecture 14.0

Tuesday, 29 September 2026 10:52

Recall.

Claim: Let E be a nonempty set of reals. Let $L < K$ be two integers such that K is an upper bound on E while L is not. Then there exists integer m with $L < m \leq K$ s.t. $\frac{m}{n}$ is an upper bound on E while $\frac{m-1}{n}$ is not for all $n \geq 1$. m is unique too.

Theorem (Existence of least upper bound).

Let E be a nonempty set of real numbers. If E has an upper bound then it has exactly one least upper bound.

Proof: Note that any nonempty $E \subseteq \mathbb{R}$ can have at most one least upper bound. So we need to show that it has at least one LUB and we are done.

Now since E has an upper bound, let it be M . Fix $n \geq 1$.

Then nM is a real number. For any real number there exists a positive integer K s.t.

$$-K < nM < K \quad (\text{This is by applying Archimedian property to } |nM|)$$

$$\Rightarrow \frac{K}{n} > M \Rightarrow \frac{K}{n} \text{ is an UB on } E \text{ because } M \text{ is one.}$$

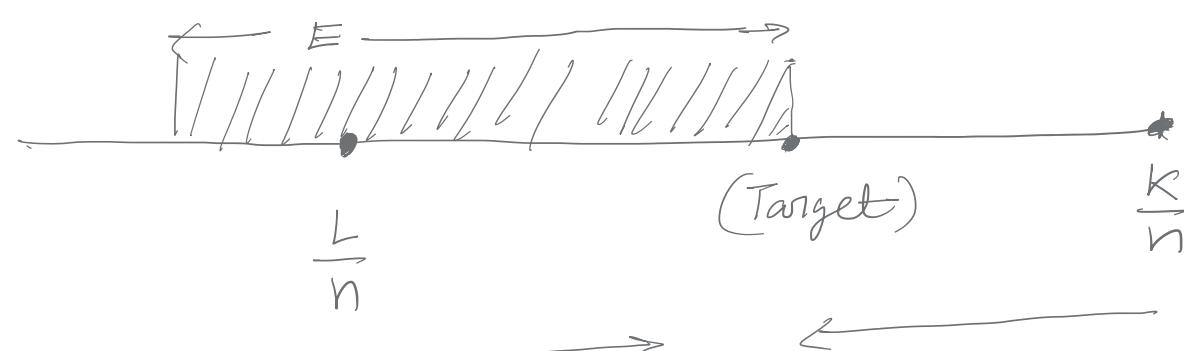
Further since E is nonempty, there is some $x_0 \in E$.

$$\text{Then again } -L' < nx_0 < L'$$

$$\text{call } L = -L' \Rightarrow x_0 > \frac{L}{n} \Rightarrow \frac{L}{n} \text{ is not an UB.}$$

$$\text{Also } \frac{L}{n} < x_0 \leq \frac{K}{n} \Rightarrow L < K.$$

$\hookrightarrow$ This is an UB.



From previous claim for each $n \in \mathbb{N}$ there exists integer m_n with $\frac{L}{n} < m_n \leq \frac{K}{n}$ s.t.

* $\frac{m_n}{n}$ is an UB on E .

* $\frac{m_n-1}{n}$ is not an UB on E .

Now consider $n, n' \geq 1$.

$$\begin{aligned} \text{Then } \frac{m_n}{n} \text{ is UB \& } \frac{m_{n'}-1}{n'} \text{ is not UB} &\Rightarrow \frac{m_{n'}-1}{n'} \leq \frac{m_n}{n} \\ &\Rightarrow \frac{m_n}{n} - \frac{m_{n'}}{n'} \geq -\frac{1}{n'} \end{aligned}$$

$$\begin{aligned} \text{K } \frac{m_{n'}}{n'} \text{ is UB \& } \frac{m_n-1}{n} \text{ is not UB} &\Rightarrow \frac{m_n-1}{n} \leq \frac{m_{n'}}{n'} \\ &\Rightarrow \frac{m_n}{n} - \frac{m_{n'}}{n'} \leq \frac{1}{n} \end{aligned}$$

let us fix $\varepsilon > 0$. Choose N s.t. $\varepsilon N > 1$ or $\frac{1}{N} < \varepsilon$.

Then $\forall n, n' \geq N$ then

$$\frac{m_n}{n} - \frac{m_{n'}}{n'} \leq \frac{1}{n} \leq \frac{1}{N} \quad \& \quad \frac{m_n}{n} - \frac{m_{n'}}{n'} \geq -\frac{1}{n'} \geq -\frac{1}{N}$$

$$\Rightarrow \left| \frac{m_n}{n} - \frac{m_{n'}}{n'} \right| \leq \frac{1}{N} < \varepsilon$$

$$\Rightarrow \left(\frac{m_n}{n} \right)_{n \geq 1} \text{ is a Cauchy sequence and } \lim_{n \rightarrow \infty} \frac{m_n}{n} \text{ is a real number.}$$

$$\text{let } S = \lim_{n \rightarrow \infty} \frac{m_n}{n} = \lim_{n \rightarrow \infty} \frac{m_n-1}{n}.$$

Now we want to show that S is a LUB.

* UB: let $x \in E$. Since $\frac{m_n}{n}$ is an UB on $E \Rightarrow$

$$x \leq \frac{m_n}{n} \Rightarrow \lim_{n \rightarrow \infty} x \leq S$$

$$\Rightarrow x \leq S \quad \forall x \in E \Rightarrow S \text{ is an UB on } E.$$

* LUB: let P be any other UB on E . since $\frac{m_n-1}{n}$ is not an UB

$$\exists x \in E \text{ s.t. } \frac{m_n-1}{n} < x \leq P$$

$$\Rightarrow \frac{m_n-1}{n} \leq P \quad (\text{strict inequality implies nonstrict})$$

$$\Rightarrow S \leq P \quad \text{For any upper bound } P.$$

This implies we found a least upper bound and it is unique.

Supremum: Let $E \subseteq \mathbb{R}$.

① If E is nonempty and has an upper bound then

$$\sup(E) := \text{LUB on } E.$$

② If E is nonempty and doesn't have an upper bound then

$$\sup(E) := +\infty.$$

③ If E is empty then $\sup(E) := -\infty$.

$\sup(E)$ is called supremum of E and is always defined.

Remark: $+\infty, -\infty$ are not real numbers, they are just symbols.

Algebra doesn't work on them.

§ Lecture 14.1

Wednesday, 30 September 2026

22:58

Claim: There exists a positive real number x such that $x^2=2$.

Proof: Consider a set

$$E = \{ y \in \mathbb{R} : y^2 < 2 \}.$$

Note that 2 is an upper bound on E , i.e. $\forall y \in E \quad y \leq 2$.

(Otherwise $\exists y \in E$ s.t. $y > 2 \Rightarrow y^2 > 4 > 2$. This can't happen).

Also $1 \in E$ as $1^2 < 2$.

Then $\sup(E) := \text{LUB on } E$ exists. Let $x = \sup(E) > 0$ (As $1 \in E$)

Assume ① $x^2 < 2$. Then for $0 < \varepsilon < 1$

$$(x+\varepsilon)^2 = x^2 + \varepsilon^2 + 2x\varepsilon \leq x^2 + \varepsilon + 4\varepsilon \quad \left(\begin{array}{l} \text{using } 2 \text{ is an UB} \\ \text{so } x \leq 2, \text{ \& } \varepsilon^2 \leq \varepsilon \end{array} \right)$$

$$\Rightarrow (x+\varepsilon)^2 \leq x^2 + 5\varepsilon$$

$$\text{Choose } \varepsilon = \frac{2-x^2}{10} > 0 \quad \text{as } x^2 < 2$$

$$\left\{ \begin{array}{l} \leq \frac{1}{5} \\ < 1 \end{array} \right. \quad \text{as } x^2 \geq 0$$

$$\Rightarrow x^2 + 5\varepsilon = \frac{x^2}{2} + 1 < 2$$

$$\Rightarrow \exists 0 < \varepsilon < 1 \text{ s.t. } (x+\varepsilon)^2 < 2.$$

$$\Rightarrow x+\varepsilon \in E \Rightarrow x \text{ cannot be an UB.}$$

[A contradiction]

Assume 2: $x^2 > 2$, let $0 < \varepsilon < 1$

$$(x-\varepsilon)^2 = x^2 + \varepsilon^2 - 2x\varepsilon \geq x^2 + 0 - 4\varepsilon$$

$$\text{choose } \varepsilon = \frac{x^2-2}{8} \Rightarrow x^2 - 4\varepsilon = \frac{x^2}{2} + 1 > 2$$

That is $\exists 0 < \varepsilon < 1$ s.t. $(x-\varepsilon)^2 > 2$

$$\Rightarrow (x-\varepsilon) > y \quad \forall y \quad \text{otherwise } \exists y \in E \text{ s.t. } (x-\varepsilon) < y$$

$$\Rightarrow (x-\varepsilon)^2 < y^2 < 2$$

(contradiction)

$\Rightarrow (x-\varepsilon)$ is an upper bound on E . But it can't be for as x is LUB.

$$\Rightarrow x = \sup(E) \text{ s.t. } x^2 = 2.$$

Remark: $\mathbb{Q}$ do not follow the least upper bound property.

If they did then there will be a rational y s.t. $y^2=2$.

which we showed is impossible.

One can similarly define infimum. [Greatest lower bound and so on]

Real exponentiation:

→ Exponentiation of real by naturals.

→ Inductive $x^0=1$, x^n is defined inductively.

$$x^{n+1} = x^n \cdot x \quad [x \in \mathbb{R}].$$

→ Exponentiation of real by an integer.

$$\rightarrow \text{Let } x \neq 0, \quad x^{-n} = \frac{1}{x^n}.$$

→ Exponentiation of real by rationals.

→ Let $x \geq 0$. $n \geq 1$ Then

$$x^{\frac{1}{n}} := \sup \{ y \in \mathbb{R} : y \geq 0 \text{ and } y^n \leq x \}$$

$$(x^{\frac{1}{2}} = \sqrt{x}) \quad \left[\begin{array}{l} \text{\& } n\text{th root of positive} \\ \text{real number } x \end{array} \right].$$

For $n=2$ we showed existence of real x such that $x^2=2$.

Similarly the existence of $x^{\frac{1}{n}}$ can be proven.

$$E_n := \{ y \in \mathbb{R} : y \geq 0 \text{ and } y^n \leq x \}$$

Proof strategy: $\left\{ \begin{array}{l} x \leq 1 : \text{Find upper bound} \\ x > 1 : \text{Find upper bound} \end{array} \right\} \rightarrow \text{LUB will exist and that is the } n\text{th root.}$

Then we can talk about properties

$$\textcircled{1} \quad x > y \Leftrightarrow x^{\frac{1}{n}} > y^{\frac{1}{n}} \quad n \geq 1.$$

$$\textcircled{2} \quad (x^{\frac{1}{n}})^{\frac{1}{m}} = x^{\frac{1}{nm}} \quad \left(\text{First prove } x^{\frac{1}{n}} = y \Leftrightarrow x = y^n \right)$$

Then one can define $x^{a/b}$ as well. ($x > 0$, $b > 0$, a any integer).

$$x^{a/b} = (x^{\frac{1}{b}})^a.$$